MEROPENEM

Reserve antibiotic

Lowers valproate levels

CHEMICAL NAME (4R,5S,6S)-3-[[(3S,5S)-5-(dimethylcarbamoyl)pyrrolidin-3-yl]sulfanyl]-6-[(1R)-1-hydroxyethyl]-4-methyl-7-oxo-1-azabicyclo[3.2.0]hept-2-ene-2-carboxylic acid (trihydrate)

CLASS Carbapenem (β-lactam), broadest-spectrum reserve antibiotic

TRADE NAMES Meronem · Merrem · Meropen

THERAPEUTIC USES / INDICATIONS

- Severe infections with multidrug-resistant (ESBL) Gram-negative bacteria.
- Hospital-acquired / ventilator-associated pneumonia; sepsis.
- Complicated intra-abdominal infections; febrile neutropenia.
- Bacterial meningitis (incl. in children).

ADMINISTRATION

- **IV** bolus over 5 min or infusion over 15–30 min (extended 3 h infusions in critical care).
- Use within stability time after reconstitution.
- Antimicrobial stewardship approval often required.

DOSAGES

- **Adult:** 1 g IV every 8 h; meningitis/cystic fibrosis 2 g every 8 h.
- **Children (3 months–12 yr):** 10–20 mg/kg every 8 h; meningitis 40 mg/kg every 8 h.
- **CrCl 26–50:** every 12 h; **10–25:** half dose every 12 h; **<10:** half dose every 24 h.

PHARMACOKINETICS

- **Onset:** immediate IV. Good tissue and CSF penetration.
- Stable to renal dehydropeptidase (no cilastatin needed).
- Excreted mainly unchanged in urine. **t½:** 1 h.

PHARMACODYNAMICS

- Binds **PBPs** → inhibits cell-wall synthesis → bactericidal.
- Very broad: Gram-positive, Gram-negative incl. *Pseudomonas* and ESBL producers, anaerobes.
- Not active against MRSA, *E. faecium*, *Stenotrophomonas*, carbapenemase producers.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Diarrhoea, nausea, vomiting.
- Headache, rash, itching.
- Injection-site inflammation.

ADVERSE EFFECTS (SERIOUS)

- Seizures (lower risk than imipenem).
- *C. difficile* colitis; fungal superinfection.
- Anaphylaxis; SJS/DRESS.
- Thrombocytopenia, raised liver enzymes.

CONTRAINDICATIONS & CAUTIONS

- Hypersensitivity to carbapenems; severe (anaphylactic) allergy to any β-lactam.
- **Caution:** epilepsy/CNS disorders, renal impairment.

KEY INTERACTIONS

- Sodium valproate: levels fall dramatically (60–100% within 24 h) → seizures — avoid combination.
- **Probenecid:** ↑ meropenem levels.
- **Warfarin:** ↑ INR.

NURSING MANAGEMENT

- Confirm allergy history and stewardship approval.
- Take cultures before first dose.
- Check for valproate therapy.
- Monitor temperature, WBC, CRP, renal function, LFTs.
- Watch for seizures, diarrhoea, oral thrush.

PATIENT EDUCATION

- Report rash, fits, severe diarrhoea.
- Tell staff if you take valproate for epilepsy.

KEY POINTS

- Carbapenem; reserve for resistant Gram-negatives (ESBL).
- Destroys valproate levels.
- 1 g IV every 8 h; adjust in renal impairment.
- Safer seizure profile than imipenem.

SOURCES CHECKED **[148]** Meropenem 1 g powder for solution for injection or infusion (UK SmPC) **[149]** Meropenem, IV (US label) **[150]** Meropenem, IV (US label, Amneal Pharmaceuticals LLC)

Full references p. 432–457